

Acoustics and Hydrodynamics of the Drop Impact: Two Modes of Sound Packet Emission

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Abstract—Acoustic signals have been studied by coordinated optical and acoustic methods in two modes of a free-falling water drop merging with water. Only the primary sound packet is observed in the intrusive mode at a low drop velocity when a cavity is formed with a delay in the colored wake of the inflowing liquid. The influence of the drop velocity on the parameters of the primary signal was traced in the experiments. The cavity begins to form from the moment of initial contact when the drop velocity is high. The flow pattern becomes more complicated, the drop substance is distributed over the surface of the emerging cavity and growing crown in the form of individual fibers forming linear and reticular structures. The falling section of the primary signal is constituted by a group of pulses with a complex spectral composition. Further, after a pause, one or more low-frequency sound packets are recorded. The spectral portrait of the secondary packets is related to the shape of the detached gas cavity. A smooth axisymmetric cavity radiates a monochromatic packet. A complex-shaped cavity transforming into a gas bubble emits a multi-frequency signal. The duration and amplitudes of the secondary sound signals depend on the rate of transformation of the separated gas cavity into a smooth spheroidal bubble. The parameters of acoustic signals change non-monotonically with increasing velocity of the merging drop.

Keywords: drop, cavity, splash, sound packets, flow modes

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INTRODUCTION

Studies of flows created by a fallen drop in a liquid, which began in the late 19th century [1], were supplemented when sound receivers were invented by recording sound signals in the air environment and, after some delay, also in the water environment. The acoustic signals include a high-frequency primary pulse and delayed sound packets emitted by gas bubbles [2]. The correspondence between the volume-oscillation spectra of the bubble and the sound packet has been established experimentally [3].

Current studies of the drop impact—a sequence of atomic–molecular, physical and hydrodynamic processes—are driven by the scientific content of the subject and by the variety and importance of practical applications. Drops participate in the exchange of matter between the lithosphere, hydrosphere, and atmosphere. Droplets falling on solid ground [4] or into water produce fine spray that is carried by the wind to form aerosols containing minerals. Mineral residues at high altitudes serve as condensation centers required for the formation of individual drops and clouds [5]. Droplets also carry viruses and bacteria that cause plant and animal diseases [6]. The development of viral epizootics, epidemics, and global pan-

demics has intensified the study of droplet transport mechanisms in the atmosphere [7].

Search for the causes of weather anomalies and determination of the effects of anthropogenic factors reinvigorated investigation into the total precipitation and associated acoustic noise in the World Ocean [8]. Algorithms for the localization of rainfall areas and estimation of swell and wind speed in remote regions were gradually developed [9].

The difficulty of work in marine conditions has stimulated laboratory tests of sound generation by individual drops and groups of drops falling in a liquid to simulate rain [10]. Sound packets can generate both rapidly falling raindrops and snowflakes slowly gliding down into water [11]. The results of measurements of acoustic noise in the ocean and its laboratory simulation formed the basis for a new scientific discipline—remote acoustic and radar meteorology [12]. Acoustic systems for remotely determining the precipitation intensity began to be used also on land [13]. Laboratory studies of drop impact acoustics continue in order to improve the accuracy of determining meteorological parameters based on the results of acoustic measurements.

The calculation of sound generation by spherical bubbles is based on the isothermal perfect fluid theory. In experiments, sound-emitting gas cavities resulting from the interaction of thin flows and capillary waves have both regular axisymmetric [5] and complex angular shapes [14].

The momentum of the impact of a spherical drop in contact with a flat liquid surface was calculated under the assumption of supersonic velocity of the interface of merging liquids [15]. Visualization of the flow pattern by X-rays during the first microseconds of the drop contact [16] and by optical methods during the first fractions of milliseconds of the coalescence process [17] showed that thin fast radial jets broke the interface of the liquids into separate segments. The smooth liquid surface at the top of the forming splash became rough [18].

As the drop velocity decreases, the flow structure is radically reorganized and the cavity forms in the intrusive mode with some delay after the drop liquid has entered the target medium [19]. The properties of acoustic signals over the range of intrusive coalescence conditions have not been studied previously. The purpose of this work is to study the influence of a free-falling drop merging with the target fluid on the parameters of acoustic signals.

1. PARAMETERIZATION OF FLOWS

Experiments were planned and the results are described subject to the properties of full solutions to the system of fundamental fluid mechanics equations derived under the condition of compatibility [20] with physically justified boundary conditions.

The main parameter of the state and flow of a medium is the total energy including all its components—kinetic energy, potential energy, and internal energy. The complex structure of liquids comprises various atomic-molecular formations, such as individual ions, complexes, clusters, clathrates, voids with solitary atoms, contact and free surfaces, and internal energy, which are distributed unevenly both on the surface and in the liquid bulk [21]. Internal energy is rapidly converted to other forms when the flow pattern is rearranged. In fact, continuous rearrangement of the distribution pattern of atoms and molecules makes liquids and gases flow. Energy in a fluid is transferred by flows at a local velocity, by waves at a group velocity, by slow diffusion-dissipative processes, and it is rapidly transformed during the reorganization of the microstructure [22].

The full system of equations of motion, including the equations of state for the Gibbs potential G and its derivative, liquid density ρ , which are not given here for brevity, makes it possible to take into account all the mechanisms of energy transfer in droplet flows [22]. Its physical quantities and dimension factors

define the set of physical quantities of the coalescence process and flow evolution.

The defining physical parameters of the problem include densities of contacting media—of the drop, ρ_d ; of the air, ρ_a ; and of target fluid, ρ_t (hereinafter $\rho_{a,d,t}$); kinematic viscosity $\nu_{d,a,t}$ and dynamic viscosity $\mu_{d,a,t}$; the surface tension coefficients of contacting media, full σ_d^a , σ_t^a and normalized to density, $\gamma_d^a = \sigma_d^a/\rho_d$, $\gamma_t^a = \sigma_t^a/\rho_t$, cm^3/s^2 ; the diffusion coefficient of pigment coloring the droplet in the target fluid, κ ; equivalent diameter D ; surface area S_d ; volume V ; mass M ; free fall height of the drop H and its velocity U at the time of initial contact; available potential surface energy (APSE), $E_\sigma = \sigma S_d$; kinetic energy $E_k = MU^2/2$; and potential energy in a gravitational field with gravitational acceleration g .

The share of the APSE E_σ distributed in the near-surface spherical layer with a thickness of the order of the molecular cluster size $\delta_\sigma \sim 10^{-6}$ cm and mass, $M_\sigma = \rho V_\sigma$ under the experimental conditions was about 1% of the kinetic energy of the drop, E_d . At the same time, the density of surface energy, $W_\sigma = E_\sigma/M_\sigma$ (M_σ is the volume of the near-surface spherical layer with thickness of the order of the molecular cluster, $\delta_\sigma \sim 10^{-6}$ cm) was high, the ratio of the densities of energy components being around $\text{Wr}_k^\sigma = E_\sigma M/E_k M_\sigma \sim 1000$.

The duration of the transfer process for the kinetic energy E_k and momentum $\mathbf{p}_k = M \cdot \mathbf{v}$ of the merging drop, $\Delta t = D/U$, which is determined by its diameter D and contact velocity U is several milliseconds. The annihilation of the APSE contacting liquids is much faster, $\Delta \tau_\sigma = \delta_\sigma/U \sim 10^{-8}$ s. The brevity of the processes of APSE transformation into other forms at the circular boundary of the liquid coalescence region plays a decisive role in the formation of the thin structure of flows and wave generation [17, 18].

The ratios of physical quantities form eigen temporal and spatial scales whose sets determine the requirements for the measurement technique in terms of choosing the size of the flow observation region, spatial and spatial and temporal resolution of instruments [8, 14, 16]. Sets of basic dimensionless parameters are used for the parameterization of flows including the numbers of Reynolds $\text{Re}_d = UD/\nu_d$, Froude $\text{Fr}_d = U^2/(gD)$, Weber $\text{We}_d = U^2 D/\gamma_d$, Bond $\text{Bo} = gD^2/\gamma_d^a$, Ohnesorge $\text{Oh}_d = \nu_d/\sqrt{\gamma_d^a D}$, and Schmidt $\text{Sc} = \nu_t/\kappa_d^m$. The magnitude of fast atomic-molecular processes of the APSE transformation is characterized by

the ratios of the energy components, $Er_k^\sigma = \frac{E_\sigma}{E_k}$, and of their densities, $Wr_k^\sigma = \frac{E_\sigma M}{E_k M_\sigma}$. The great number of dimensionless parameters reflect the complexity of the spatiotemporal structure of the investigated flows where several competing processes of jet and wave formation are at work simultaneously.

There are two types of waves during the drop impact flows including (1) mechanical gravitational-capillary waves and (2) sound waves [1, 3, 21]. The frequency ω and wave number k of gravitational-capillary waves are linked by the dispersion relation [7]:

$$2k(\omega^2 k_l - gk^2 - \sigma k^4 + 2i\omega v k_l(3k^2 - k_l^2)) - (k_l^2 + k^2)(\omega^2 - gk - \sigma k^3 + 2i\omega v k^2) = 0. \quad (1)$$

The regular solution of Eq. (1) uniformly becomes Thomson's dispersion relation for capillary waves in ideal fluid. Ligaments, thin jets affecting the formation of gas bubbles, correspond to the singular solution of the operator $\hat{L} = i\omega + v\Delta$ of the system of equations [20] with the wave number k_l [14].

The dispersion equation for sound waves in a homogeneous compressible fluid includes the wave and multiple singular factor generated by the operator $\hat{L}$ [22]:

$$\left(\mathbf{k}^2 \left(1 - \frac{i\omega \tilde{v}}{c_s^2} \right) - \frac{\omega^2}{c_s^2} \right) (\omega + i v \mathbf{k}^2)^2 = 0. \quad (2)$$

Here c_s is the speed of sound. In drop flows, capillary waves are observed in case of abrupt rearrangement of the flow pattern and acoustic waves are produced when gas bubbles are detached.

2. EXPERIMENTAL SETUP

The experiments were carried out on the Free Surface Effects (FSE) stand of a unique HPC IPMech RAS research facility [32] that included a transparent basin of $1.5 \times 0.4 \times 0.6 \text{ m}^3$ with optical portholes. A measuring hydrophone (bandwidth 0.002 to 100 kHz, sensitivity 30 mV/Pa) was immersed in the basin.

A droplet dispenser, ReyLab Xenos RH-1000 lighting 1 kW spotlights, and Optronis MultiLED sources with a luminous flux of 7700 lm were placed above the basin. Flows were recorded with Optronis CR3000x2 and FastCam 1024PCI high-speed video cameras. The experiment control, data collection, and transfer to the computer were carried out through the synchronization interface of signals from the video cameras and the hydrophone with resolution up to 10^{-7} s (4 channels, digit capacity 12 and clock frequency 10 MHz). A calibrated marker was recorded for automatic determination of the size of the test objects (photos of the stand are given in [21]).

The basin and the drop dispenser were filled with partially degassed tap water. In the experiments, the falling drop interrupted a photodetector whose signal with an adjustable delay triggered the interface and computer that recorded the signals. The data were processed using specialized mathematical software.

3. RESULTS AND DISCUSSION

3.1. Primary Acoustic Signals in the Intrusive Coalescence Mode

A sample from the video record of merging with water of a 0.42 cm drop of dilute ink solution falling freely from a height of 2 cm is shown in Fig. 1. The degree of ink dilution varied in the experiments more than by 1000 times, no qualitative influence of ink concentration on the properties of the studied phenomena was observed.

Eigen-oscillation modes are excited in the drop as it detaches from the dispenser, their interference causing rapid changes in its shape. In this experiment, the central part of the approaching drop is spherical and its head part is conical with a smoothed top (Fig. 1a). The free surfaces at the interface are eliminated when the mixing liquids comes into contact and the drop begins to flow into the target fluid bulk where it forms a colored area with a rounded head (Fig. 1b, $t = 3.75 \text{ ms}$). The mode in which the total free surface of the target fluid and of the drop remainder continue to be convex for a long time is called intrusive mode [19].

The inflowing drop remainder stretches vertically and continues to oscillate, its bottom flattening out. A group of three circular capillary waves $\lambda = 0.38 \text{ mm}$ long run upward along the lateral surface of the drop from the interface. It breaks the lateral surface of the vertically elongated drop, where the waves are clearly visible in the form of a sequence of circular belts (Fig. 1b). In addition, irregularities of the lower edge of the invading fluid can be seen in an enlarged image. They are irregular protrusions, depressions, and tufts of protruding thin colored fibers. The drop inflow process in the intrusive mode lasts from 3 to 9 ms depending on the contact velocity, which was determined from measurements of the drop displacement.

Then the lower edge of the forming intrusion is smoothed out. The oscillatory motions in the drop remainder continue throughout the coalescence process, the drop bottom becoming conical at $t = 7.75 \text{ ms}$ (Fig. 1c). The fluid surface remains convex until the drop is completely submerged (Fig. 1d, $t = 13 \text{ ms}$).

A conical cavity begins to form rapidly after the drop inflow ends in the interval of $13 < t < 21 \text{ ms}$. The rapidly growing cavity (the bottom velocity at the beginning, $u_c = 0.68 \text{ m/s}$, formation time $\Delta t_c = 8 \text{ ms}$) pushes down the spot of colored liquid (Fig. 1d, $t = 25 \text{ ms}$), which transforms into a classical ring vortex

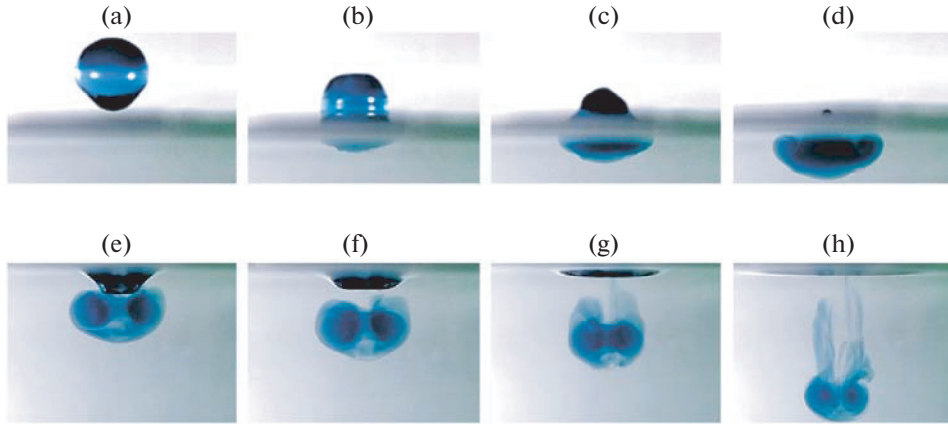


Fig. 1. Coalescence of a free-falling ink-dyed drop with water in the intrusive mode ($D = 0.42$ mm, $U = 0.6$ m/s): (a) approaching drop, (b–h) $t = 3.75, 7.75, 11, 25, 29.25, 37.25, 62.5$ ms.

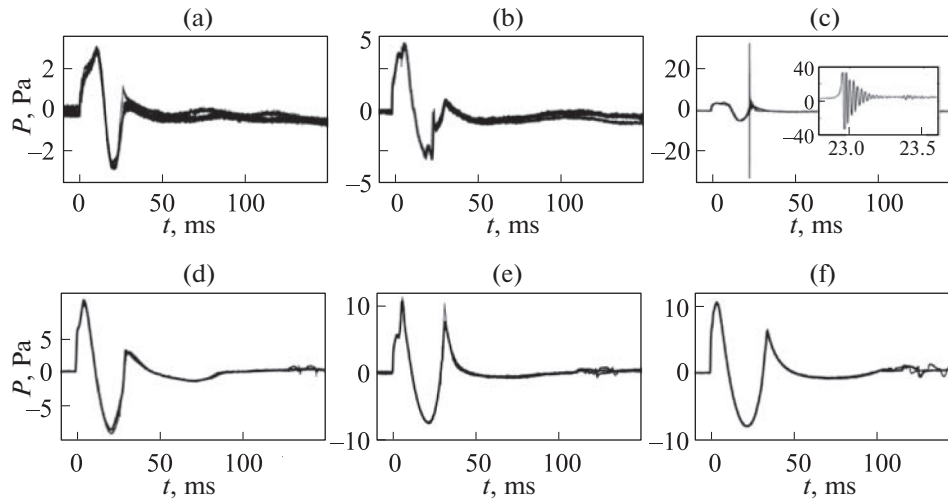


Fig. 2. Phonograms of acoustic pressure in the intrusive mode of drop coalescence ($D = 5.0$ mm): (a–f) $U = 0.63, 0.77, 0.88, 1.08, 1.32, 1.40$ m/s (inset in Fig. 2c is the sweep of a resonance packet with frequency $f = 41$ kHz).

with a pronounced toroidal core (outer diameter $d_r = 0.6$ cm, inner diameter $d_i = 0.36$ cm, Fig. 1e, $t = 25.35$ ms).

Next, acted upon by buoyancy and surface tension forces, the cavity shrinks towards the free surface while the ring that gained speed continues to submerge because the colored fluid is slightly heavier than pure water (Fig. 1f, $t = 29.25$ ms).

The ink captured by the cavity surface is carried to the near-surface layer, where it formed a thin colored spot (Fig. 1g, $t = 37.25$ ms) whose diameter monotonically increases and thickness decreases. The submerging ring leaves a colored fibrous wake (Fig. 1h, $t = 62.5$ ms). There is a streamer (thin splash) in the image of the free surface (in the frontal projection) at the collapse phase in the center of the cavity. Its top does not reach the level of the unperturbed fluid sur-

face and is not shown in the lateral projection/vertical plane.

The time dependences of acoustic pressure in the intrusive mode shown in Fig. 2 differ markedly from traditional phonograms [5, 15, 21, 27]. Signals in three independent experiments synchronized by initial contact time were placed there in each chart. The conditions of the acoustic experiments in the intrusive mode are summarized in Table 1. Since all the experiments were conducted with drops of the same diameter, the Bond and Ohnesorge numbers did not change and were $Bo = 3.36$ and $Oh = 1.7 \times 10^{-3}$, respectively.

It follows from the comparison of the curves in Fig. 2 that the higher the velocity, the more accurately the shapes of the primary acoustic signals are reproduced. The discrepancy of the curves in Figs. 2a, 2b is a consequence of chance variation in the shape of a detached oscillating drop. In only one case in the

whole series of these experiments (and only in one of the three realizations shown in Fig. 2c), a tone signal with a frequency of $f = 41$ kHz (the equivalent diameter of the spherical bubble according to the formula [4] $d_g = 0.16$ mm) was recorded.

The spikes and kinks in the phonograms indicate the variation in the properties of the emitting source and possible excitation of oscillations of small bubbles (Figs. 2b, 2e in the areas of primary and secondary growth of acoustic pressure). Small oscillations of the signal at $t > 100$ ms in Figs. 2d–2f are due to the development of small-scale perturbations of the free surface during the splash growth phase [18].

The comparison of optical images in Fig. 1 and of phonograms in Fig. 2 allows to trace the connection of hydrodynamic processes with acoustic signals.

The initial contact of a drop, at which liquids begin to merge and the drop momentum and kinetic energy are transferred, is accompanied by the elimination of the free surface of merging liquids, by releasing APSE during the calculated time $\Delta t = \delta_\sigma / U \sim 10^{-9} - 10^{-8}$ s, and by the formation of the pressure pulse front lasting $3.4 < t_m < 9.8$ ms.

The maximum slope of the pressure front is recorded at the initial stage and then the increment rate decreases, which is consistent with the APSE release rate variation. As the drop head submerges, the surfaces of both the drop and of the target fluid are eliminated. At the same time, the APSE release rate is higher than that of the drop bottom submergence when the fluid contact ring—the outer boundary of the coalescence region—shrinks towards the center of the flow. At this phase, a part of the motion energy is spent to replenish the growing APSE surface. The maximum values of acoustic pressure P_m , which were achieved during the time t_m , and of the signal rise slope $G = P_m/t_m$ in these experiments are given in Table 1.

The comparison of the above data shows that the maximum pressure perturbation increases non-monotonically with increasing drop velocity (the linear regression curve has the form $P_m = 6.7U + 23.5$ [Pa], $[U] = \text{m/s}$), and the duration of the leading front decreases, $t_m = 12.2 - 6.4U$, $[t] = \text{ms}$. The signal slope is approximated by the expression $G = 10.4U - 2.9$, $[G] = \text{Pa/ms}$ and generally increases as the drop velocity rises. In one of the experiments (at $U = 1.25$ m/s), its value was several times higher than the interpolated value indicating a strong dependence of the signal on the shape of the drop surface at the time of initial contact (pointed or flattened) and on the amount of released APSE.

The duration of the signal decay phase, $\tau = 11U + 3.2$, $[t] = \text{ms}$, increases with drop velocity growth owing to long inertial cavity deepening. The duration of the first phase of the process at the zero level (width of the pressure growth peak) decreases as

Table 1. The acoustic experiment conditions in the intrusive mode of drop coalescence

U , m/s	Re	Fr	We	E_k , erg	Er_k^σ	$W_{rk}^\sigma \times 10^{-4}$
0.63	3150	08.10	27.2	129.9	0.44	22
0.77	3850	12.10	40.6	194.1	0.30	15
0.88	4400	15.80	53.0	253.4	0.23	11
1.00	5000	20.42	68.5	327.2	0.17	8.8
1.08	5400	23.80	79.9	381.7	0.15	7.5
1.17	5850	27.90	93.8	447.9	0.13	6.4
1.25	6250	31.90	107.0	511.3	0.11	5.5
1.33	6650	36.10	121.1	578.8	0.099	0.45
1.40	7000	40.00	134.2	641.4	0.089	0.42

the velocity $T_0 = 17.7 - 4.8U$, $[T_0] = \text{ms}$, increases due to a shorter drop coalescence time.

Then the cavity bottom begins to approach rapidly the free surface as the cavity shrinks. A small circular protrusion is formed on the fluid surface around the cavity, which increases pressure on the hydrophone. As the ring spreads out, the pressure returns to zero. The duration of the primary signal, including the pressure rise phase caused by the formation of the crown, grows monotonically, $T = 20.5U + 19.9$, $[T] = \text{ms}$.

The pressure variations at $t > 15$ ms are due to hydrodynamic processes, initially slow and increasingly high-frequency ones at 100 ms, which are manifested in subtle perturbations of the free surface [18].

3.2. Drop Impact Acoustics in the Impact Mode of Cavity and Splash Formation

As drop velocity rises and the flow goes to the splash formation mode, the drop spreading pattern changes radically. The cavity delineated by the ring crown was observed from the time of initial contact (Fig. 3a). The cavity bottom is flat at the initial stage indicating that the momentum and energy of the drop are uniformly distributed over the entire spreading surface (Fig. 3b). Spikes protrude from the tops of the teeth on the edge of the crown. Sequences of small droplets (spatter) fly out from the ends of the spikes. The drop substance is distributed in fibers across the cavity bottom and crown walls [18].

The spikes on the crown edge gradually shrink into rounded droplets (Fig. 3c). Circular capillary waves run down the walls of the crown to the base of the cavity. The cavity bottom rounds off and becomes spherical, the edge of the falling crown being smoothed out (Fig. 3d).

Further, the cavity starts to collapse and its bottom is flattened again by buoyancy and surface tension forces. The cavity bottom is roughened by thin near-surface flows at the splash formation phase (Fig. 3e).

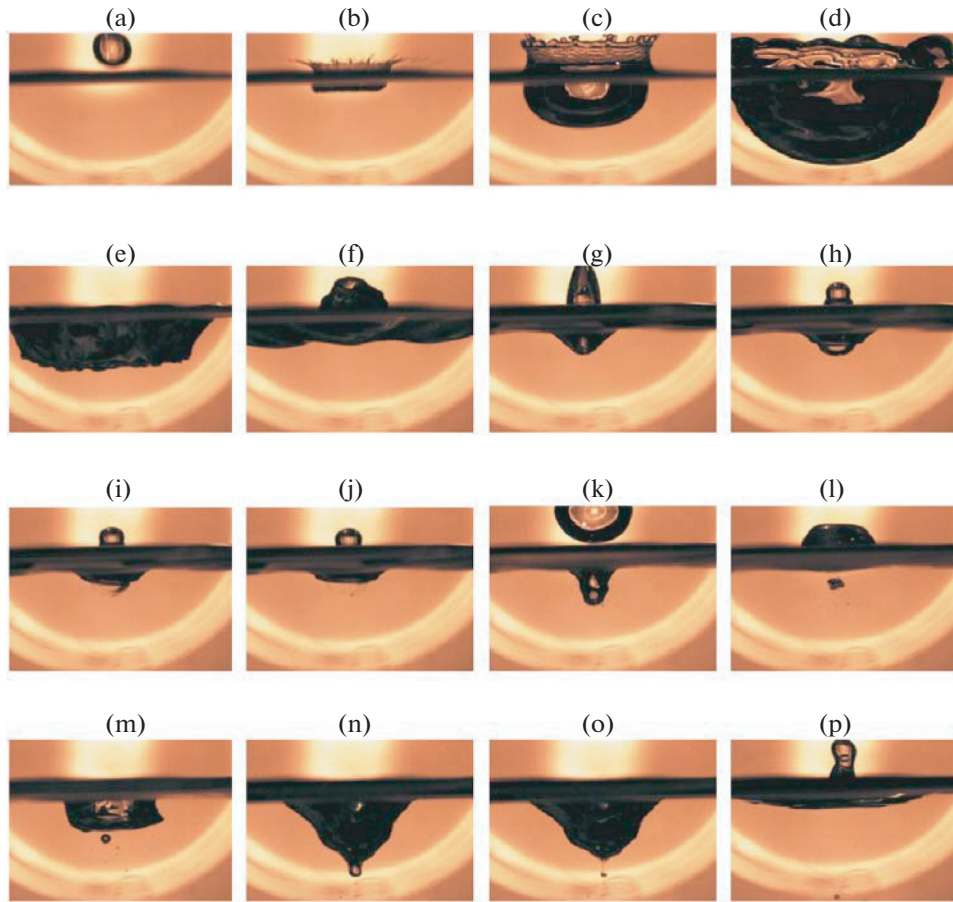


Fig. 3. Evolution of the flow pattern as a water drop spreads in water in the splash formation mode ($D = 5$ mm, $U = 3.57$ m/s): (a) approaching drop, $t = -0.5$ ms; (b–p) $t = 1.25, 4.25, 31.00, 44.75, 55.75, 144.25, 149.25, 149.5, 149.75, 160.75, 169.00, 184.25, 200.75, 201.25, 219.75$ ms.

The cavity being smoothed out continues to widen with the head of the growing splash appearing at its center (Fig. 3f). Having reached its maximum height, the splash begins to subside, its broad base pushing through the surface of the fluid whose center starts to move faster than the splash top falls. The cavity becomes conical and the base of the splash is sharpened (Fig. 3g).

Next comes a complicated stage of the flow evolution. The cavity bottom moves faster than the splash in the $139 < t < 143$ ms interval. And the splash base thins out and detaches from the target fluid surface. The cavity bottom stops after a few microseconds and begins to rise returning to its initial position. The contact of the cavity bottom with the pointed base of the splash generates short internal waves that can be seen in Fig. 3h (see [14] for details). In the process, the splash liquid does not flow deep into the target fluid, but rapidly fills the conical gap between the splash and the cavity whose wall is covered with capillary waves (only the left part of the cavity is filled in Fig. 3i, the right part is dry and remains dark). The next frame shows a “bubble wake”—a cloud of small gas bubbles

reproducing the cross-section of the cavity being filled (Fig. 3j).

The splash that continues to submerge forms a new complex-shaped cavity (Fig 3k) from which an angular gas cavity breaks away (Fig 3l). The remnant of the merging drop can be seen on the fluid surface. It touches the cavity walls with its lateral surface in the vicinity of the equator. The APSE released during the elimination of the free surfaces forms thin flows that distort the shape of the walls of the growing cavity (Fig. 3m). The transformation of the angular gas cavity into a smooth spheroidal one is accompanied by the emission of a monochromatic sound packet whose frequency agrees with calculations [2, 3].

The movements of small gas bubbles in Figs. 3k–3n visualize a complex pattern of flows near the cavity walls stretching the thin gas cavity at its base (Fig 3n). The next frames shows that the top of the cavity has broken off and is moving into the fluid (Fig. 3o). Such cavities that are formed breaking a thin bridge usually emit a monochromatic sound packet [3]. Once the second cavity is filled, quite a few “silent bubbles” remain in the fluid. One of them is captured by a grow-

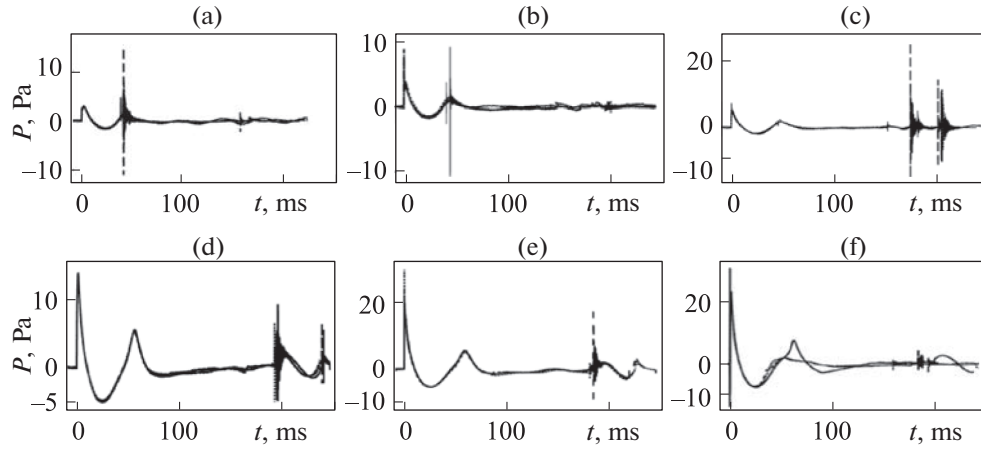


Fig. 4. Phonograms of acoustic pressure in the impact mode of drop coalescence ($D = 5.0$ mm): (a–f) $U = 1.9, 2.4, 2.8, 3.7, 4.4$, and 4.8 m/s.

ing streamer, a new narrow protruding jet in the center of the cavity (Fig. 3p).

The conditions of this series of acoustic experiments are summarized in Table 2 (the Bond and Ohnesorge numbers were $Bo = 3.36$ and $Oh = 1.7 \times 10^{-3}$, respectively).

The phonograms shown in Fig. 4 include one or more low-frequency sound packets lagging behind the primary signal. The data from three independent experiments shown in the same chart combine well in all experiments except for those in Fig. 4f at the highest drop velocity ($U = 4.8$ m/s). The interaction of the drop with the air medium begins to manifest itself in the trajectory there. It depends on its shape and modes of oscillations at the moment of detachment.

The formation of resonance sound packets is synchronized with two time intervals during which the flow is restructured. When the drop contact velocity $U < 2.5$ m/s, secondary packets are formed with a delay $\Delta t = 40$ to 60 ms when a splash begins to form in the flow and to rise above the undisturbed fluid level (Fig. 3f). The fluid surface near the top of the forming splash is deformed by thin components of intense convergent near-surface flows.

The structure of the primary packet that includes two peaks separated by a smooth trough is maintained over the entire range of drop velocities, but the signal parameters listed in Table 3 differ markedly when the drop velocity changes. The magnitude of the primary packet peak pressure P_s in the impact mode appears to be lower than in the intrusive mode and grows irregularly as the drop velocity increases. The primary signal rise time t_m reaches the minimum at the boundary between the modes and grows further with increasing contact velocity, which indicates a change in the mechanism of interaction between merging liquids.

The slope of the primary momentum G grows at first as velocity increases, then it reaches a plateau that is interrupted by a sharp increase almost by an order of magnitude ($G = 4000$ Pa/ms at $U = 4.4$ m/s) and then drops back down sharply again. Primary momentum

Table 2. The acoustic experiment conditions in the splash formation mode

U , m/s	Re	Fr	We	E_k , erg	Er_k^σ	Wr_k^σ
2.0	10000	82	274	1309	0.044	2200
2.4	12000	118	395	1885	0.030	1300
2.8	14000	160	537	2566	0.022	1100
3.1	15500	196	658	3145	0.018	900
3.4	17000	236	792	3783	0.015	750
3.7	18500	279	938	4480	0.012	600
4.0	20000	327	1096	5236	0.011	550
4.2	21000	360	1208	5773	0.010	500
4.4	22000	395	1326	6336	0.009	450
4.6	23000	432	1449	6925	0.008	400
4.8	24000	470	1578	7540	0.007	350

Table 3. The parameters of the primary sound packet in the splash formation mode

U , m/s	P_s , Pa	t_m , ms	G , Pa/ms	τ , ms	T_0 , ms	T , ms
1.9	2.8	8	240	22.2	10.2	54
2.4	8.7	9	833	24.9	9.8	56
2.8	4.9	5.5	725	25.0	10.8	62.3
3.7	6.8	8	810	24.5	9.2	68.1
4.4	23.9	4	4000	27	10	71.8
4.8	12.5	13.5	425	24.8	10.1	82.5

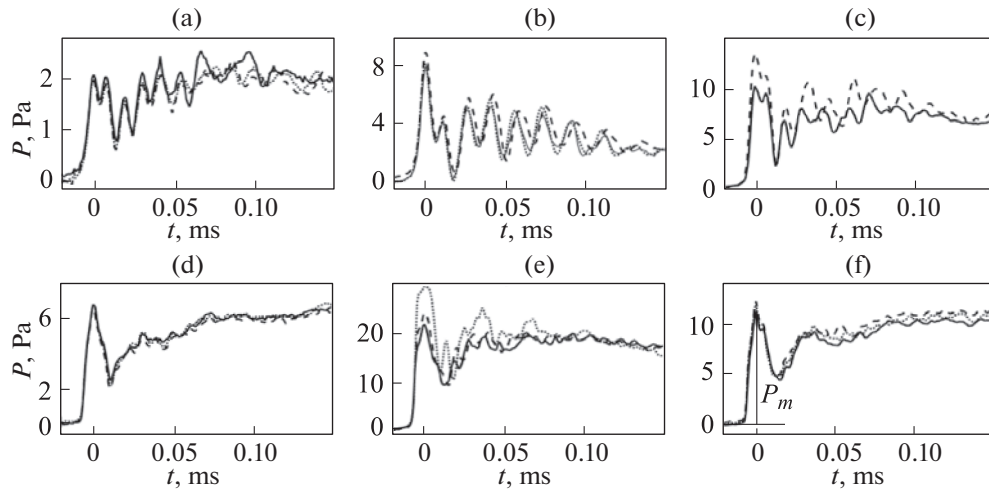


Fig. 5. Phonograms of acoustic pressure in the impact mode of drop coalescence ($D = 5.0$ mm): (a–f) $U = 1.9, 2.4, 2.8, 3.7, 4.4$, and 4.8 m/s.

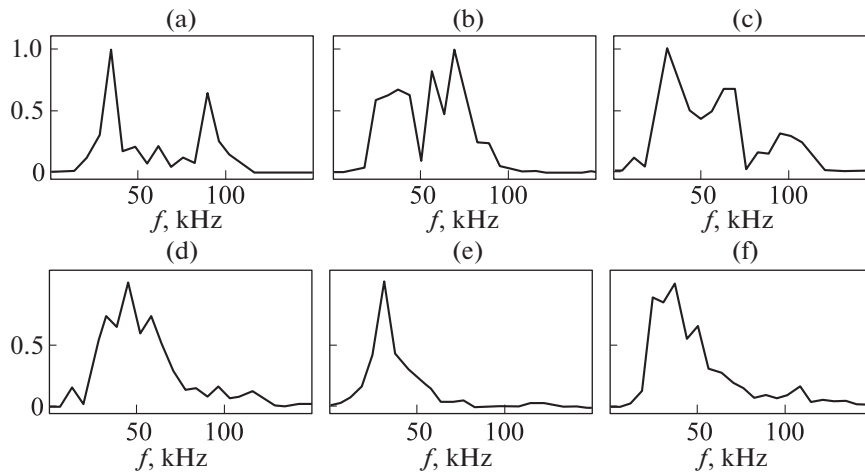


Fig. 6. Spectral portraits of the phonograms of decaying parts of impact momenta in relative units ($D = 5.0$ mm) of drop velocity U and frequencies of discrete components f : (a–f) $U = 1.9, 2.4, 2.8, 3.7, 4.4$, and 4.8 m/s.

width, T_0 , and the duration of the signal decay phase, τ , fluctuate near the mean value, and the total duration of the primary acoustic perturbation T grows with increasing drop energy.

The series of stretched sections of phonograms illustrates an important difference between the complex declining section of the primary signal in the splash mode and the smooth section in the intrusive mode. The signals of complex spectral composition depend non-monotonically on velocity there (Figs. 5 and 6).

The primary momentum is accompanied in the phonogram of the striking impact signal by a series of sound packets of increasing amplitude (Figs. 5a, 5d, 5f) or decreasing amplitude (Figs. 5b, 5c) with pronounced (Figs. 5a, 5b) or small (Figs. 5d, 5f) variations of parameters. The conditions of the experiments

being maintained, complex signals lose consistency (Fig. 5a). The signals are slightly shifted in frequency (Fig. 5b) or in amplitude of individual components (Figs. 5c, 5e). As the drop velocity increases, while the uncertainty of the behavior of the signal that can diminish (Fig. 5e) or rise (Fig. 5f) is maintained, the amplitude of secondary oscillations decreases. No resonance sound packets were observed at a contact velocity greater than $U > 4.6$ m/s.

The spectral portraits of signals in Fig. 5 express discrete components, two of them initially with frequencies $f = 34.2$ and 89.0 kHz (Fig. 6a). As the drop velocity rises, the signal structure becomes more complex and three peaks are shown in Fig. 6b, $f = 37.9, 60.1, 69.6$ kHz and in Fig. 6c, $f = 31.4, 69.1, 94.2$ kHz, the frequency difference between which grows with increasing velocity. When the drop velocity becomes

Table 4. The parameters of the resonance sound packet in the splash formation mode

U , m/s	P , Pa	t_m , ms	G , Pa/m	τ , ms	T_0 , ms	T , ms
1.98	26.24	8.15	2.67	9.96	15.70	33.10
2.42	30.79	7.12	4.32	2.97	13.46	38.71
2.80	30.05	4.55	6.60	11.82	12.90	33.66
3.13	28.95	6.79	4.26	11.21	12.90	40.39
3.42	32.57	5.29	6.15	16.53	1.78	43.76
3.70	30.67	1.18	26.08	18.77	1.79	42.08
3.96	30.84	4.11	7.51	17.47	11.78	44.32
4.20	31.55	6.36	4.96	15.36	11.78	46.56
4.43	34.37	3.41	10.09	19.04	11.24	51.17

The resonance wave packet was not formed for 4.64 and 4.85 m/s.

greater than $U > 3.1$ m/s, only one discrete component remains in the signal, $f = 44.8$ kHz (Fig. 6d) and $f = 31.6$ kHz (Figs. 6e, 6f).

The parameters of the resonance signals (of lower frequency compared to the primary packets) are listed in Table 4. Here the maximum perturbation of pressure P and the slope of signal G grow non-monotonically with increasing drop velocity U while the duration of the leading front, t_m , drops.

The duration of the signal decay area, τ , grows monotonically, which is associated with an increase in the duration of inertial deepening of the cavity. The total duration of the resonance sound packet grows with increasing drop energy.

The mechanisms of the formation of the primary (impact) signal and subsequent discrete components have not been identified at present, the process of sound generation by drops falling into a liquid needs further study.

4. CONCLUSIONS

Detailed recording of sound signals and visualization of the fine structure of flows were carried out for the first time over a wide range of parameters including intrusive and impact modes of a free-falling drop merging with water in a deep basin.

The primary sound packet in the intrusive mode, when the cavity is formed with delay after the end of the drop liquid inflow process, includes an increasing front, a smooth decay of the signal to negative values and its slight rise. Slow pressure decline and subsequent rise are due to the hydrostatic influence of the evolving flow components such as the cavity and small central and circular protrusions. The interpolation value of the signal was noticeably exceeded in some experiments against the background of a monotonic change in the signal parameters as the drop velocity increased.

In the impact mode of splash formation, when the cavity and crown begin to form from the initial contact of the drop with the target fluid, the impact momentum is followed with some delay by one or more sound packets.

The primary momentum front is steeper there, high-frequency oscillations being registered in the decaying part of the primary signal. The longer duration of the initial perturbation compared to the intrusive mode is owing to an increase in the cavity depth with increasing drop velocity.

The secondary sound packet is recorded during the crown growth phase and/or at the break of the declining splash. A group of capillary waves is generated at the cavity wall when the splash detaches. The spreading liquid of the splash that continues to submerge overlaps the wave crests. Air remaining in the wave troughs generates gas cavities of complex shape, which break off forming a thin channel.

Next gas cavities break off from the complex-shaped cavity formed during the coalescence of the drop that had previously escaped from the top of the splash. The duration of sound and the spectral composition of the signal depend on the shape of the cavity that can be both rounded and flat, angular, or with some protrusions and depressions. The gas cavities are transformed into spheroidal gas bubbles in the process of emission of acoustic packets.

The processes of relatively slow transfer of kinetic energy (characteristic time was several milliseconds in these experiments) and rapid release of APSE (fractions of microseconds) during elimination of the free surfaces of merging liquids play an important role in the formation of the drop impact flow pattern.

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CONFLICT OF INTEREST

The author of this work declares that he has no conflicts of interest.

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